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SUB : AI_ASSISTANT_COADING

ASSIGNMENT : 7.3

Task 1: Fixing Syntax Errors

```
def add(a, b)
```

```
    return a + b
```

```
result = add(5, 3)
```

```
print(result)
```

File "task1.py", line 1

```
def add(a, b)
```

```
    ^
```

SyntaxError: expected ':'

> "Fix this Python syntax error: def add(a, b) return a + b"

python

```
def add(a, b):
```

```
    return a + b
```

```
result = add(5, 3)
```

```
print(result)
```

Task 2: Debugging Logic Errors in Loops

```
count = 0
```

```
while count < 5:
```

```
    print(count)
```

```
    # Forgot to increment count!
```

> "Why does this loop run infinitely? count = 0 while count < 5: print(count)"

```
python
```

```
count = 0
```

```
while count < 5:
```

```
    print(count)
```

```
    count += 1 # Fixed: added increment
```

Task 3: Handling Runtime Errors (Division by Zero)

Code:

```
python
```

```
def divide(a, b):
```

```
    return a / b
```

```
print(divide(10, 2))
```

```
print(divide(10, 0)) # Crashes here!
```

Error :

5.0

Traceback (most recent call last):

File "task3.py", line 5, in <module>

print(divide(10, 0))

File "task3.py", line 2, in divide

return a / b

ZeroDivisionError: division by zero

"Add error handling to this division function to prevent crashes when dividing by zero"

Code:

python

def divide(a, b):

try:

return a / b

except ZeroDivisionError:

return "Error: Cannot divide by zero!"

print(divide(10, 2)) # Output: 5.0

print(divide(10, 0)) # Output: Error: Cannot divide by zero!

Task 4: Debugging Class Definition Errors

python

```
class Student:
```

```
    def __init__(name, age):
```

```
        name = name
```

```
        age = age
```

```
    def display(self):
```

```
        print(f"Name: {self.name}, Age: {self.age}")
```

```
s = Student("Rahul", 20)
```

```
s.display()
```

Error :

Traceback (most recent call last):

File "task4.py", line 9, in <module>

```
s = Student("Rahul", 20)
```

TypeError: Student.__init__() takes 2 positional arguments but 3 were given

"Fix this Python class error: class Student: def __init__(name, age): name = name age = age"

Corrected Code:

```
class Student:
```

```
    def __init__(self, name, age): # Fixed: added 'self' as first parameter
```

```
self.name = name # Fixed: assign to instance variables
```

```
self.age = age
```

```
def display(self):
```

```
    print(f"Name: {self.name}, Age: {self.age}")
```

```
s = Student("Rahul", 20)
```

```
s.display() # Output: Name: Rahul, Age: 20
```

Task 5: Resolving Index Errors in Lists

Code:

```
fruits = ["apple", "banana", "cherry"]
```

```
print(fruits[3]) # Trying to access 4th element (index 3)
```

ERROR:

Traceback (most recent call last):

File "task5.py", line 2, in <module>

```
    print(fruits[3])
```

IndexError: list index out of range

AI Prompt Used:

```
> "How to safely access list elements in Python without getting IndexError?"
```

Corrected Code

python

```
fruits = ["apple", "banana", "cherry"]
```

```
index = 3
```

```
if index < len(fruits):
```

```
    print(fruits[index])
```

```
else:
```

```
    print(f"Error: Index {index} is out of range (valid: 0-{len(fruits)-1})")
```

Output: Error: Index 3 is out of range (valid: 0-2)

Alternative Fix (using try-except):

```
fruits = ["apple", "banana", "cherry"]
```

```
index = 3
```

Method 2: Use exception handling

```
try:
```

```
    print(fruits[index])
```

```
except IndexError:
```

```
    print(f"Invalid index {index}!")
```

